

%nprocshared=64
%mem=320GB
%chk=PbRhaC12C12T2.chk
opt=loose freq apfd/gen scrf=(solvent=water) pseudo=read scf=xqc

PbRhaC12C12T2

0 1

Pb	-0.49927595	-4.34204683	1.34599140
O	2.12775861	-2.95695229	1.70239846
O	-0.62101611	-2.00685318	0.93847349
O	2.14846695	-4.38929556	-0.59614328
C	5.65732978	-3.34440324	-0.05398687
O	5.28912124	-1.99694188	0.27385195
C	4.27818691	-1.93888614	1.25169287
C	2.92751017	-2.40754219	0.66430895
C	3.19820463	-3.47087318	-0.41391659
C	4.43532255	-4.26986814	-0.04311730
O	-0.18804663	-3.12921218	-0.91543458
C	-0.41796543	-2.04586816	-0.30595406
C	-0.44913407	-0.79021146	-1.14126099
C	-0.52146997	0.54107480	-0.41166981
O	0.55850345	0.67668111	0.54988588
C	-1.79881599	0.81349633	0.35947358
C	-3.05190347	0.70726323	-0.50708706
C	-4.21729169	1.50299379	0.06692779
C	-5.52454284	1.29217341	-0.68553360
C	-6.60592337	2.28371088	-0.27793997
C	-7.94815950	2.04078587	-0.95459058
C	1.78480319	0.88443538	0.07491984
O	2.06863624	0.89354699	-1.10398242
C	2.77613685	1.14479091	1.17601874
C	4.13011425	0.45806396	0.95341150
O	4.23397813	-0.67910270	1.83052586
C	6.77721802	-3.84434814	0.83806512
O	4.67908381	-5.31337681	-0.96619558
C	5.28382830	1.40747591	1.24358718
C	6.65795263	0.77710582	1.09634698
C	7.78663671	1.76509297	1.35401901
C	9.16576470	1.12519238	1.27130161
C	10.30067902	2.10484531	1.53670590
C	11.68140684	1.46536674	1.46023048
C	12.80708428	2.45613177	1.72937358
O	0.17620189	-4.18124520	4.09772902
O	-2.50420768	-3.59905355	2.44561897
O	-0.72218217	-6.57412002	3.51631269
C	-3.37284153	-5.46902174	5.89546459
O	-2.59633071	-4.38132177	6.42631064
C	-1.81805823	-3.78201270	5.41616086
C	-0.64017364	-4.69782316	5.14323413

C	-1.17018676	-6.11026071	4.78179838
C	-2.69059689	-6.12067005	4.68583947
O	-2.77369809	-4.24421921	0.35771952
C	-3.23686930	-3.76410340	1.42601729
C	-4.70129053	-3.39505453	1.51064956
C	-4.99102579	-2.00007454	2.05333472
O	-4.56058123	-1.96170358	3.43366654
C	-6.47802665	-1.69417611	1.97040457
C	-6.88217683	-0.35869880	2.57517477
C	-8.36756283	-0.08014171	2.39472359
C	-8.81165862	1.24646759	2.99341318
C	-10.28855437	1.53704651	2.76557492
C	-10.72485279	2.89634476	3.29642696
C	-3.86141647	-0.90101272	3.84738082
O	-3.66235134	0.09385080	3.18794383
C	-3.26016737	-1.14227540	5.20029933
C	-1.76445862	-1.47481611	4.98893692
O	-1.34794125	-2.54226979	5.84562614
C	-4.77429678	-5.00428817	5.54429785
O	-3.18923199	-7.44216053	4.59394764
C	-0.84585943	-0.29784498	5.22941038
C	0.60122819	-0.62762944	4.89313689
C	1.56712077	0.53358403	5.08071973
C	3.02011304	0.09716440	4.94866333
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H	-0.37902777	1.32649001	-1.16105632
H	5.99937986	-3.29173357	-1.09148650
H	4.51514785	-2.60994787	2.08798287
H	2.40535582	-1.56158180	0.20457572
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H	-2.84256643	1.07150900	-1.52200087
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H	-4.34861249	1.24303527	1.12255398
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H	6.74182304	-0.06876054	1.78603350
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H	7.72387718	2.59245282	0.63448997
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H	-8.61086285	-0.09411456	1.32490113
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H	-8.21175871	2.05928370	2.56442302
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H	-3.37240289	-0.25054196	5.81962288
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H	3.16912862	-0.42259331	3.99544531
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H	5.66343959	0.04200149	5.72021939
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H	1.96794784	-2.24749271	2.33456656
C	14.19147555	1.78666073	1.64649216
H	14.33166362	1.37725657	0.66790361
H	14.25298563	1.00347185	2.37294359
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H	16.24787597	2.37069226	1.85899361
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H	15.15136097	3.23576058	2.90893563
C	7.91355840	1.44472345	4.94158480
H	8.11649651	0.74550486	5.72568268
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C	8.86748589	2.64722704	5.06651453
H	9.81512944	2.39683032	4.63736049
H	8.45244407	3.48697202	4.54934082
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H	-10.70981891	1.85102712	-0.97996166
C	-11.35924373	3.89364528	-0.83507669
H	-11.63878031	4.47818907	-1.68658586
H	-12.22811115	3.42276158	-0.42490849
H	-10.91551216	4.52842603	-0.09679782
C	-12.61818992	4.55991327	3.58908710
H	-12.44548835	4.59378872	4.64451432
H	-12.04214149	5.32302371	3.10874278
C	-14.11417357	4.79124031	3.30603104
H	-14.51939518	5.44993294	4.04550346
H	-14.23022543	5.22885746	2.33653468
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C H O 0
6-311+g(2d,p)

Pb	0	
S	4	1.00
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	46.757232559	0.56961959130D-02
	20.746462696	-0.21374063831
	14.610796419	0.40502620616
S	2	1.00
	20.222637612	-0.83541883299D-01
	6.4767324865	0.97910892388
S	1	1.00
	1.6600600927	1.0000000
S	1	1.00
	0.80431655001	1.0000000
S	1	1.00
	0.22627039020	1.0000000
S	1	1.00
	0.84014530665D-01	1.0000000
P	3	1.00
	15.189102118	0.61952303583
	14.693144415	-0.72498497086
	6.8705890048	0.37680007984
P	3	1.00
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	1.2209125119	0.46058131862
	0.63367559815	0.19367655397
P	1	1.00
	0.28202837058	1.0000000
P	1	1.00
	0.11333375666	1.0000000
P	1	1.00
	0.43948707430D-01	1.0000000
D	6	1.00
	61.315369628	0.33870800787D-03
	12.372195840	0.13788683942D-01
	6.9254944983	-0.75979608103D-01
	2.3319539939	0.28113784298
	1.2108730003	0.44474512269
	0.60090478506	0.35326874351
D	1	1.00
	0.28135869813	1.0000000
D	1	1.00
	0.11500000000	1.0000000
F	1	1.00
	1.0000000	1.0000000
F	1	1.00
	0.2896200	1.0000000

PB 0

PB-ECP 3 60

f potential

2

2 3.88751200 12.20989200

2 3.81196300 16.19029100

s-f potential

4

2 12.29630300 281.28549900

2 8.63263400 62.52021700

2 3.88751200 -12.20989200

2 3.81196300 -16.19029100

p-f potential

6

2 10.24179000 72.27689700

2 8.92417600 144.59108300

2 6.58134200 4.75869300

2 6.25540300 9.94062100

2 3.88751200 -12.20989200

2 3.81196300 -16.19029100

d-f potential

6

2 7.75433600 35.84850700

2 7.72028100 53.72434200

2 4.97026400 10.11525600

2 4.56378900 14.83373100

2 3.88751200 -12.20989200

2 3.81196300 -16.19029100